

Signal Processing

Exercise 1 – Fourier Series

Summer Semester 2026

- Only applied on **periodic** signals:
Periodic signals can be represented as a sum of sine and cosine waves.
- The amplitude spectrum of a Fourier series consists of discrete values.
- (non-periodic signals → Fourier transform).

Definition of the Fourier series

- Superposition of sine and cosine oscillations are defined as **real Fourier series**
→ Trigonometric series expansion.
- Occurring frequencies are integer multiples (harmonics) $n \cdot \omega_0$ of the fundamental angular frequency ω_0 .

$$y(t) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cdot \cos(\omega_0 n t) + b_n \cdot \sin(\omega_0 n t)] \quad \omega_0 = 2\pi f_0 = \frac{2\pi}{T_0}$$

$y(t)$: Signal to be analyzed

a_n, b_n : Amplitudes of the sine and cosine waves

ω_0 : Fundamental angular frequency

- Complex Fourier series

$$y(t) = \sum_{-\infty}^{\infty} c_n \cdot e^{j\omega_0 n t}$$

Terminology

Harmonic:	$1 \cdot f_0$	→	1st harmonic (fundamental component)
	$n \cdot f_0$	→	n -th harmonic

Calculation of the coefficients

Coefficient of the DC component $a_0 = \frac{2}{T} \int_T y(t) dt$

Real coefficients

$$a_n = \frac{2}{T} \int_T y(t) \cdot \cos(\omega_0 n t) dt$$

$$b_n = \frac{2}{T} \int_T y(t) \cdot \sin(\omega_0 n t) dt$$

Complex coefficients

$$c_n = \frac{1}{T} \int_T y(t) \cdot e^{-j\omega_0 n t} dt$$

NOTE: The more abruptly the signal changes, the **a)** greater values take the magnitudes $|c_n|$ of higher frequency components and **b)** the higher the order of n becomes up to which calculation needs to be performed.

Conversion of the real and complex Fourier coefficients for real signals

For **real signals**, the negative frequency components c_{-n} and positive frequency components c_n are related by the complex conjugate.

1. Real to complex coefficients

$$c_0 = \frac{a_0}{2}; \quad c_n = \frac{1}{2}(a_n - jb_n); \quad c_{-n} = c_n^* = \frac{1}{2}(a_n + jb_n)$$

c_0 : DC component of the signal

2. Complex to real coefficients

$$a_0 = 2c_0; \quad a_n = c_n + c_{-n} = c_n + c_n^*; \quad b_n = j(c_n - c_{-n}) = j(c_n - c_n^*)$$

Properties of the Fourier series

Even signals (symmetry with respect to y-axis)

$$\rightarrow b_n = 0$$

- $y(t)$ is periodic in T_0
- $y(-t) = y(t)$

Odd signals (point symmetry)

$$\rightarrow a_n = 0$$

- $y(t)$ is periodic in T_0
- With respect to zero: $y(-t) = -y(t)$
- In case of a DC offset: $y(-t) - c_0 = -y(t) + c_0$